

Durham College	
Course	INFT1100: Network Systems and Components
Lab 1	Configuring IP Address

Output Section: You will only submit this page. Please don't submit any instructions	
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Output 1: 3 pings with IP and subnet mask

```

Administrator: Command Prompt

C:\Windows\system32>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=3ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Windows\system32>ping 205.211.182.80

Pinging 205.211.182.80 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 205.211.182.80:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Windows\system32>ping durhamcollege.ca
Ping request could not find host durhamcollege.ca. Please check the name and try again.

C:\Windows\system32>

```

Output 2: 3 pings with IP, subnet mask and default gateway

```
Administrator: Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=97ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=21ms TTL=64

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 97ms, Average = 30ms

C:\Windows\system32>ping 205.211.182.80

Pinging 205.211.182.80 with 32 bytes of data:
Reply from 192.168.2.19: Destination host unreachable.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 205.211.182.80:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),

C:\Windows\system32>ping durhamcollege.ca
Ping request could not find host durhamcollege.ca. Please check the name and try again.

C:\Windows\system32>
```

Output 3: 3 pings with IP, subnet mask, default gateway and DNS server

```
Select Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=1ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64
Reply from 192.168.2.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Windows\system32>ping 205.211.182.80

Pinging 205.211.182.80 with 32 bytes of data:
Reply from 205.211.182.80: bytes=32 time=25ms TTL=48
Reply from 205.211.182.80: bytes=32 time=27ms TTL=48
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48

Ping statistics for 205.211.182.80:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 27ms, Average = 26ms

C:\Windows\system32>ping durhamcollege.ca

Pinging durhamcollege.ca [205.211.182.80] with 32 bytes of data:
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48
Reply from 205.211.182.80: bytes=32 time=26ms TTL=48

Ping statistics for 205.211.182.80:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 26ms, Maximum = 26ms, Average = 26ms

C:\Windows\system32>
```

Output 4: Summary of the activity

1. With IP and subnet mask you can communicate with the default gateway
2. The role of the default gateway is to provide a way for network traffic to leave a local network and to connect to other external networks, such as the internet.
3. The role of the DNS server is to take a domain name like durhamcollege.ca and turn it into a numerical IP address that computers use to locate each other on the network.

Output 5:

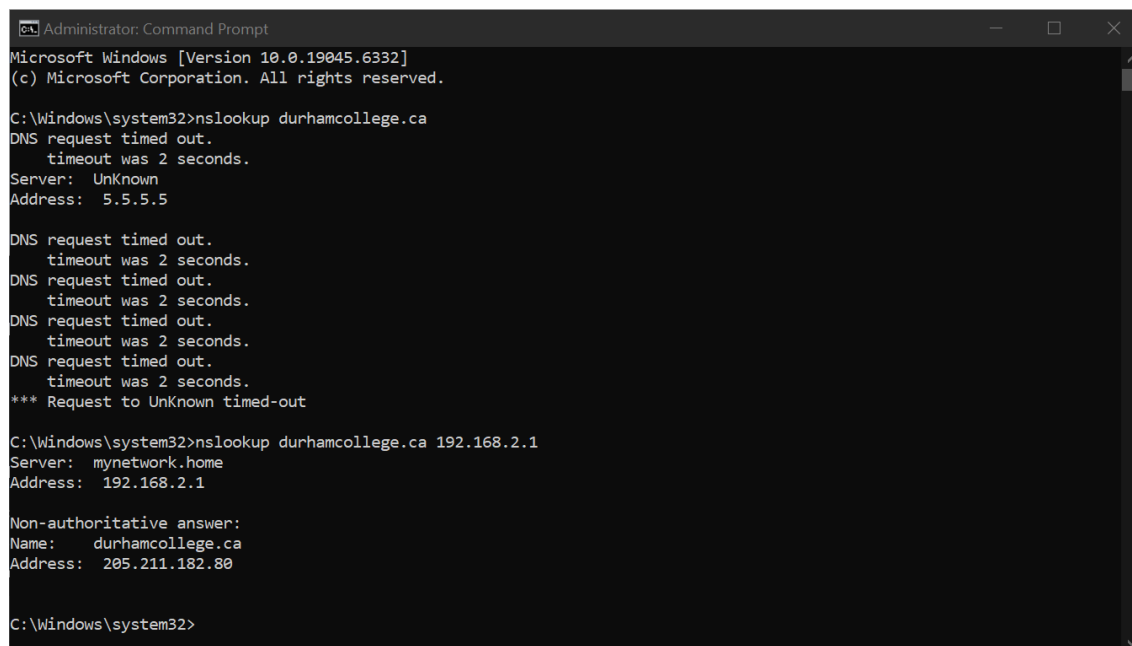
The IP address returned did not change because domain names will always result in the same IP address regardless of the DNS server used unless one DNS server has an updated IP address and another still needs to be updated. The DNS server changed because I changed the DNS server in my adapter settings from my home DNS server to Google's 8.8.8.8.

Output 6:

My home network DNS server was used to resolve the domain name durhamcollege.ca because it successfully returned a numeric IP address. The alternate DNS server will be used if the preferred DNS server fails.

The preferred DNS server failed even when the alternate DNS server is valid because nslookup uses the preferred DNS server in the network settings if executed without any arguments.

This experiment demonstrates that you must use the DNS server IP address when executing nslookup as an argument as the command will always use the preferred DNS server if you do not explicitly tell it to use another DNS server.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>nslookup durhamcollege.ca
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: 5.5.5.5

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out

C:\Windows\system32>nslookup durhamcollege.ca 192.168.2.1
Server: mynetwork.home
Address: 192.168.2.1

Non-authoritative answer:
Name:    durhamcollege.ca
Address: 205.211.182.80

C:\Windows\system32>
```